

Software Engineering Intro

Condensed Study Notes

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Course Contents

This course covers key areas of software development:

- **Software Design:** Focuses on structuring software effectively.
 - Software architecture: The high-level structure of a system.
 - Design patterns: Reusable solutions to common design problems.
 - OO analysis & Design: Object-Oriented approaches to understanding and building software.
 - Design for re-use: Creating software components that can be used in multiple projects.
- **Using APIs:** Interacting with existing software components.
 - API programming: Writing code to use Application Programming Interfaces.
 - Class browsers and Related tools: Tools for inspecting and understanding code structure.
 - Component based computing: Building systems from independent, interchangeable parts.
- **Software tools and Environment:** Tools that aid the development process.
 - Requirements analysis and design modeling tools: Software for capturing and visualizing requirements and designs.
 - Testing tools: Software for verifying program correctness.
 - Tool integration mechanisms: Ways to connect different development tools.

The Term Software

Software is a collection of instructions that guides a system to take inputs, process them, and produce desired outputs according to user specifications for function and performance.

It also encompasses documentation, like user manuals, which aid in understanding the software system.

Today's software is comprised of

Software is made up of several key components:

- **Instructions (Computer Programs):** These are the core commands that tell the computer what to do to achieve desired functions and performance.
- **Data Structures:** These are organized ways to store and manage information, allowing programs to effectively manipulate data.
- **Documents:** These describe how the software operates and how to use it. Some features might be optional based on user choices.

Software is also developed with specific hardware and operating system environments in mind, known as the **platform**. Therefore, software descriptions include its capabilities and the platform specifications

needed to run it.

Additionally, the raw content mentions related concepts that are part of the overall software development lifecycle:

- **Source Code:** Programs written by developers in programming languages (e.g., Java, Python).
- **Executable:** Compiled code that can run independently of the source code.
- **Design Documents:** Various design specifications like input/output design, UML, and architecture.
- **Operations:** The expected functions and features of the software.
- **System Manuals:** Guides for users and technical documentation.
- **Installation:** The process of setting up software on a user's system, dependent on hardware and OS.
- **Implementation:** Integrating the software into an organization's existing systems after installation.

The Term Engineering

Engineering is defined as the profession and activity of designing structures like roads and bridges, or machinery and equipment, based on acquired knowledge.

At its core, engineering involves achieving goals within associated constraints. An engineer manages this process through measurement, action, and controlled procedures, by identifying needs and securing resources to produce a product or service.

More broadly, engineering can be understood as 'to make, achieve, or get through contrivance or guile'.

- **Contrivance:** A device, often mechanical, invented for a specific function.
- **Guile:** Cunning or deceit used to solve problems, particularly those not strictly mechanical in nature.

Ultimately, engineering is characterized as a verb – an active approach to problem-solving, rather than just a field of study.

Source vs. Executable in Visual Basic

This section provides a practical guide on distinguishing between source code and executable files within the Visual Basic environment.

- **Source Code:** The human-readable instructions written by a programmer. It's like a recipe, detailing the steps and ingredients needed.
- **Executable File:** The program that a computer can directly run. It's the finished cake, ready to be served, created from the recipe.

Visual Basic allows developers to write source code, which is then compiled into an executable file that users can interact with.

The Term Software Engineering

Software Engineering applies scientific knowledge and systematic, disciplined, quantifiable approaches to building software. It focuses on creating cost-effective solutions to practical problems, satisfying user needs through design, construction, and testing.

This approach is particularly important for larger and complex software systems, emphasizing best practices and engineering processes in development, operation, and maintenance. It is essentially the application of engineering principles to software.

Key aspects include:

- Analyzing user needs.
- Designing, constructing, and testing end-user applications.
- Applying engineering principles to software development.
- Focusing on best practices and processes for complex systems.

Computer Science VS Software Engineering

Computer Science (CS) focuses on the theoretical and mathematical aspects of how computers function.

Software Engineering (SE) studies the practical process of building software systems, encompassing areas like project management, architecture, risk, and quality assurance.

Both fields teach programming fundamentals and computer science basics, making either suitable for becoming a software developer.

Choose Computer Science if you enjoy math and logic, or aim for specialized areas like AI, machine learning, security, or graphics.

Choose Software Engineering for a hands-on approach to understanding the complete software development and maintenance lifecycle.

OR

Software Engineering is the application of scientific knowledge and methods to create practical, cost-effective solutions for software's design, construction, operation, and maintenance.

Its goal is to produce fault-free software that meets user needs, delivered on time and within budget. The term itself signifies a movement towards more systematic software development, encompassing various methods and techniques.

The key differences are

Computer Science (CS) offers more elective choices in advanced topics. Students can explore areas like security, computer vision, machine learning, and database management.

Software Engineering (SE) has more required courses in core SE fundamentals, including software testing, design principles, and specifying software requirements.

Importance of Software

Software is a primary driving force in modern society.

Its key roles include:

- **Business Engine:** Drives decision-making processes.
- **Scientific & Engineering Tool:** Forms the basis for investigations and problem-solving.
- **Embedded System Component:** Integrated into diverse systems like transportation, healthcare, telecommunications, and entertainment.
- **Pervasive Life Impact:** Affects nearly all aspects of daily activities, commerce, and culture.

As software's importance grows, the community focuses on developing technologies to build high-quality programs more easily, quickly, and affordably.